My understanding and the decisions I'd like your input on.

The setup: three DMAs; hydraulics assumed calibrated; first-order decay with kb and kw. For point two, I shouldn't say TOC, DBP and biofilm are 'not considered'; rather, I don't model them as separate species, but their effect is lumped into kb. Is this understanding correct?
The inlet is measured, so I input it directly as a boundary.
Monitors: three inlets as boundary, five for calibration, two held out for validation — ten in total. Is this setting reasonable?

The method:
I sample (kb, kw) from prior ranges, simulate, and compare modelled to observed — and instead of one best fit I get a distribution and prediction intervals. That's the Monte-Carlo / GLUE ensemble; the Bayesian version I haven't studied yet. I will read the material this week.

My open questions:

do the three DMAs share a source, so a single kb?
Should kb come from a jar test and be fixed while I calibrate only kw — which I think is easier and breaks the compensation — or calibrate both together?
And does temperature need handling if the data spans seasons?

Regarding the comparison across DMAs: I initially thought that since literature indicates k_w is not shared among different materials, it would be impossible to predict one DMA from another.
I think there are two possible approaches.
The first is to calibrate on A and then predict B and C, which allows me to measure the extent of reliability degradation—the degradation itself being the outcome—and the data will reveal the transferability among DMAs without relying on subjective assumptions.

The second approach is to calibrate A, B, and C separately. By comparing the calibration results with the intrinsic characteristics of A, B, and C, we can draw conclusions about under what conditions specific kc and kb values would occur.